AYUSH YADAV

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Summary

Software Engineer with hands-on experience building AI agent systems, RAG/LLM applications, and full-stack products in Python, React, and Node.js. Strong Data Structures & Algorithms foundation (500+ problems solved), plus practical exposure to LangChain, OpenAI Agents SDK, prompt engineering, and agentic-AI evaluation pipelines from an internship benchmarking LLM-based coding agents.

Technical Skills

- **Languages:** Python, JavaScript/TypeScript, C, C++, SQL
- **CS Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks
- **AI / LLM Engineering:** RAG pipelines, LangChain, OpenAI SDK & Agents SDK, prompt engineering, vector databases, LLM evaluation, agentic AI, PyTorch, Reinforcement & Federated Learning
- **Backend & APIs:** FastAPI, Node.js, Express.js, REST APIs, Socket.io
- **Frontend:** React.js, Next.js, Zustand, Tailwind CSS
- **Cloud & Data:** AWS, MongoDB, MySQL, PostgreSQL, Prisma, Convex
- **Tools & Practices:** Git/GitHub, Linux/CLI, Docker (basics), CI/CD basics, unit & oracle-based automated testing, Agile collaboration

Experience

Software Engineering Intern — Xelron *Feb 2026 – Jul 2026*
Bengaluru, India

- Built and validated AI-agent benchmark scenarios (debugging, CLI execution, error handling) across Terminal-Bench, Finch, and Terminus, improving evaluation coverage and reliability for LLM-based coding agents.
- Designed 30+ deterministic, oracle-validated edge-case tests for terminal-based agent tasks, tightening the signal-to-noise ratio of automated evaluation pipelines.
- Analyzed AI agent execution traces to identify failure patterns and evaluation inconsistencies, feeding fixes back into benchmark design.
- Collaborated with senior engineers to refine automated evaluation pipelines, task specs, and validation logic for scalable agent benchmarking.

Personal Projects

CodeSense — AI Code Review & Evaluation Platform (GitHub) (Live Site) *May 2026*

- Built a full-stack AI agent that diagnoses bugs, generates fixes, and renders a Monaco-based diff view for developers.
- Designed a two-layer evaluation pipeline combining LLM-as-judge scoring with rule-based checks for accuracy, clarity, completeness, and hallucination risk.
- Built an analytics dashboard tracking review history, score trends, and bug-type distribution over time.
- **Tech Stack:** Next.js, TypeScript, PostgreSQL, Prisma, Clerk, Monaco Editor, Tailwind CSS

Sentinel — Privacy-Preserving Federated Reinforcement Learning (GitHub) *Aug 2026*

- Built a federated RL system (PPO/A2C, PyTorch) that detects and mitigates network anomalies across distributed gateways without sharing raw data, using a custom multi-pattern traffic simulator.
- Applied federated aggregation to merge decentralized policy updates into a global model; evaluated with F1, Precision, Recall, ROC-AUC, and reward stability across normal, congested, and attack scenarios.

Fullstack Real-Time Chat Application (GitHub) (Live Site) *Jul 2025*

- Shipped a real-time chat platform (React, Node.js, Express, MongoDB, Socket.io) with bidirectional messaging and live presence, eliminating the need for page refreshes.
- Built REST APIs and persistent data models in MongoDB; managed client state with Zustand and a responsive Tailwind/DaisyUI interface.

Education

Motilal Nehru National Institute of Technology Allahabad *2022 – 2026*
Bachelor of Technology, Computer Science and Engineering — Prayagraj, India

Achievements & Coding Profiles

- **LeetCode:** peak rating 1631; solved 500+ DSA problems; global rank 7,563 / 31,000+ in Biweekly Contest 167 (handle: Ayush_Yadav1345).
- **Codeforces:** Specialist, max rating 1401 (rank 248 of 28,431, Round 1033); **CodeChef:** rank 258 of 21,146 (Starters 192); **GeeksforGeeks:** Pupil, rating 1351.
- ALPHA grade, 'C' Certificate, National Cadet Corps, MNNIT, 2024–2025; Lance Corporal (NCC) and Coordinator, Nepathaya Committee.